

Arjun Thanneermalai Shanker

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Software Engineer | Python, LLM Applications and Machine Learning | Testing and CI/CD

SUMMARY

Software Engineering graduate (dual degree, Sunway and Lancaster Universities) with a four-month internship, available immediately and open to relocate. In Python I built the large language model (LLM) layer of KIRA+'s AI financial assistant and trained a machine learning classifier, and for two live government portals I built a Flutter front end and a Figma design system. I learn quickly, work well independantly and in a team, and wish to learn & work on AI/Machine Learning for Real Software.

EDUCATION

Lancaster University and Sunway University

Bandar Sunway, Selangor

Bachelor of Software Engineering (Hons), Dual Degree (Twinning Program)

Sep 2023 - Sep 2026

CGPA 3.26 / 4.00 | Dean's List: Feb - Apr 2025 (GPA 3.67), May - Aug 2026 (GPA 3.63) | Student Ambassador (2024, 2025)

- Related Coursework: Data Structures and Algorithms, Software Architecture and Design Patterns (A+), Software Processes (A+, DevOps and CI/CD), Concurrent Programming (Java multithreading and thread pools, CUDA labs), Operating Systems (C, Linux, processes and threads), Artificial Intelligence, Object-Oriented Programming (Java, C++), Computer Vision.

Sunway College, OSSD (CIMP), Canadian International Matriculation Programme

Jul 2022 - Jul 2023

EXPERIENCE

Century Software (Malaysia) Sdn Bhd

Petaling Jaya, Selangor

Software Engineer Intern

Jan 2026 - Apr 2026

- Implemented the Flutter front end of two live government portals for Android and iOS as a component library in code, wired it to their existing back-end APIs after analysing the database schemas, and ran pre-release and regression tests.
- As the team's only UI/UX designer, designed 500+ screens for BayarNow (bill payments for the public across Malaysia) and OSC (business licensing), on a Figma component library self-built to the Malaysian Government Design System (MYDS).
- Proposed Pay Direct, a no-login payment flow adopted over the original plan, and wrote its storyboard. Collaborated with the CEO, CFO and product manager and supervisor on requirements and presented prototypes to the board of directors.

Fireball (fireball.my)

Remote

Freelance Website Developer

Nov 2025 - Feb 2026

- Alongside full-time studies, co-built the official Malaysian website of a Korean automotive accessories brand, linking its products to the client's Shopee store for checkout. Handled all client communication and took every post-launch issue (Shopee payment problems, the site not loading) through to a fix.

PROJECTS

KIRA+: AI Financial Health Platform for Buy Now, Pay Later and Consumer Credit

2026, in progress

MAIC Nexus Challenge 2026, Fintech track | AI/ML developer role, distributed team of 5 | Python, FastAPI, SQLite, GitHub Actions

- Built its AI financial assistant's LLM layer, the system prompt, a provider-switching client that answers within 4 seconds, and an output guard that rejects any reply quoting a figure the engine did not compute and prevents giving financial advice.
- Implemented the 0-100 scoring engine (six weighted risk factors, six warning rules) and trained and evaluated a RandomForest classifier on the project's 12,000 synthetic profiles: ROC-AUC 0.923 held out, 0.921 in 5-fold cross-validation.
- Worked test-first (56 unit and integration tests), then ran the app in a browser and fixed two defects the suites missed. Built the Docker image and GitHub Actions CI/CD pipeline with a teammate, and provisioned the server it deploys to.

SivaSafe: Voice-Controlled Desktop Assistant with Facial Authentication for Elderly Use

2025 - 2026

Final Year Capstone Project, graded A | Independent project over two semesters | Python end to end, PyQt6, OpenCV, Gemini

- Designed the assistant around one Gemini session that holds the spoken conversation and calls tools in 19 capability areas, choosing one agent over the proposal's multi-agent design to cut cost and delay.
- Secured sign-in with face recognition (OpenCV YuNet detection, SFace embeddings matched by cosine similarity), a spoken password transcribed offline by a Whisper model from Hugging Face, and PBKDF2-hashed recovery codes.
- Found and fixed an API-key failover that never ran (it watched for the wrong error type), root-caused two race conditions in its messaging automation, and measured Gemini's response time at about 1.1 s for a short reply and 5.2 s for reasoning.

Local AI Assistant: Tool-Calling AI Agent

2025, personal project

- Defined the tools a Gemini 2.5 Flash agent uses to control the computer by voice and wrote the code behind each, then chained calls, feeding each result back until the request was done, with fallback across Gemini, Groq and OpenRouter.

Portfolio Site: hand-written HTML, CSS and JavaScript on GitHub Pages, with its content served live from Cloud Firestore

TECHNICAL SKILLS

Programming: Python 3.14, Java, C++, C, SQL, JavaScript, TypeScript, Dart, HTML, CSS, PowerShell, Bash

AI and Machine Learning: LLM APIs (Gemini, OpenAI, Anthropic, Groq, OpenRouter), prompt engineering, output guardrails, tool-calling AI agents, LangChain (DeepLearning.AI), Hugging Face, OpenCV, scikit-learn, pandas, NumPy

Back End and Front End: FastAPI, REST APIs, Node.js, SQLite, Cloud Firestore (NoSQL), Flutter, npm, Figma

DevOps and Tools: Git version control, GitHub Actions CI/CD, Docker (Dockerfile, CI builds), DigitalOcean, Linux over SSH

Testing and Practice: pytest, unit, integration and regression tests, test-driven development, debugging, Agile and Scrum

CERTIFICATIONS AND LANGUAGES

Microsoft Certified: Azure AI Fundamentals (AI-900), 2025 | Microsoft AI-102 Exam-Prep Training, Apr - Aug 2025

LangChain for LLM Application Development (DeepLearning.AI), 2026

Languages: English (Fluent), Malay (Fluent), Tamil (Native), Mandarin Chinese (Basic, currently studying)